Refactoring a Java Fourier-Domain Circuit Simulator for Nonlinear Transistor Support

Architecture and Implementation Guide

1 Purpose and scope

The existing simulator represents signals using selected Fourier coefficients, models linear components by complex impedances, and solves a linear system at each frequency.

The proposed extension is:

1. Assemble linear components using admittance stamps and modified nodal analysis (MNA).
2. Preserve the existing independent-frequency linear solver.
3. Introduce nonlinear device models defined by terminal currents, terminal charges, and their derivatives.
4. Solve nonlinear periodic steady states using harmonic balance.
5. Adapt the active frequency set using omitted-equation residuals and estimated voltage errors.

Do not replace nonlinear transistors with fixed small-signal impedances: that approach cannot generate harmonics or intermodulation.

This guide targets steady-state simulation. General transients, autonomous oscillators, subharmonic solutions, and chaotic behavior require additional methods.

2 Conventions

Use the Fourier convention

$$v(t) = \sum_{k \in \mathcal{K}} V_k e^{jk\omega_0 t}.$$

For real signals,

$$V_{-k} = V_k^*, \qquad V_0 \in \mathbb{R}.$$

Use current injected into a network node as positive on the right-hand side. Device terminal currents are positive into the device.

Choose a ground node and omit its voltage unknown. Always retain DC, even when all independent sources are AC.

Internally, represent the independent Fourier unknowns using real numbers:

$$x = [X_0, \Re X_1, \Im X_1, \ldots]^T.$$

Here X_k contains node voltages and any MNA branch-current unknowns. Generate negative-frequency coefficients by conjugation.

Do not store both conjugate coefficients as independent unknowns.

3 Linear component assembly

3.1 Two-terminal admittance stamp

For a component between nodes a and b,

$$y(\omega) = \frac{1}{Z(\omega)}, \qquad I_{ab} = y(V_a - V_b).$$

Stamp:

$$Y_{aa} \mathrel{+}= y, \quad Y_{ab} \mathrel{-}= y,$$
$$Y_{ba} \mathrel{-}= y, \quad Y_{bb} \mathrel{+}= y.$$

Skip entries involving the ground row or column.

Component	Admittance	DC treatment
Resistor	$1/R$	Same stamp
Capacitor	$j\omega C$	Open circuit
Inductor	$1/(j\omega L)$	Use MNA
General impedance	$1/Z(\omega)$	Handle singular limits explicitly

Do not approximate ideal shorts using arbitrarily large admittances.

3.2 Independent current source

A current source I_s directed from a to b contributes

$$S_a \mathrel{-}= I_s, \qquad S_b \mathrel{+}= I_s.$$

For periodic sources, use their Fourier coefficients at each frequency.

3.3 Modified nodal analysis

Use MNA for ideal voltage sources, inductors, and other constraints that require branch-current unknowns.

For a branch from a to b, let b denote its incidence vector: $+1$ at node a, -1 at node b, and zero elsewhere.

An ideal voltage source with unknown current I_s gives

$$\begin{bmatrix} Y & b \\ b^T & 0 \end{bmatrix} \begin{bmatrix} V \\ I_s \end{bmatrix} = \begin{bmatrix} S \\ V_s \end{bmatrix}.$$

An inductor with current I_L satisfies

$$b^T V - j\omega L I_L = 0,$$

and contributes the block

$$\begin{bmatrix} Y & b \\ b^T & -j\omega L \end{bmatrix}.$$

At DC this enforces a short circuit without division by zero.

Call the complete linear MNA matrix $A_{\mathrm{L}}(\omega)$. It need not be symmetric.

3.4 Voltage-controlled current source

For an output current from a to b,

$$I_{ab} = g(V_c - V_d),$$

stamp

$$Y_{ac} \mathrel{+}= g, \quad Y_{ad} \mathrel{-}= g,$$
$$Y_{bc} \mathrel{-}= g, \quad Y_{bd} \mathrel{+}= g.$$

This is the basic building block of small-signal transistor models.

4 Nonlinear device contract

For each device, define reduced terminal voltages

$$u = BV,$$

where B extracts voltage differences from global node voltages.

Define reduced conductive currents and charges:

$$f(u), \qquad q(u).$$

The device contributes node currents

$$i_{\text{node}} = B^T \left(f(u) + \frac{d}{dt} q(u) \right).$$

The model must also provide

$$G(u) = \frac{\partial f}{\partial u}, \qquad C(u) = \frac{\partial q}{\partial u}.$$

The mapping and current definitions must be paired correctly. Reduced currents are not necessarily currents entering the terminals used to define the voltage differences.

For a first implementation, allow $q = 0$ and $C = 0$. Add charge storage only after conductive models are validated.

Charge-based modeling is preferred to inserting an arbitrary voltage-dependent capacitance: the physical current is dq/dt.

5 Initial transistor model: Ebers–Moll BJT

For an NPN device, define

$$u = \begin{bmatrix} v_B - v_E \\ v_B - v_C \end{bmatrix}.$$

Let

$$I_F = I_{ES} \left(e^{v_{BE}/V_T} - 1 \right), \qquad I_R = I_{CS} \left(e^{v_{BC}/V_T} - 1 \right),$$

where

$$V_T = \frac{k_B T}{q_e}.$$

Terminal currents positive into the transistor are

$$i_C = \alpha_F I_F - I_R,$$
$$i_E = -I_F + \alpha_R I_R,$$
$$i_B = (1 - \alpha_F) I_F + (1 - \alpha_R) I_R.$$

Verify

$$i_B + i_C + i_E = 0.$$

For this voltage mapping, use the reduced current vector

$$f(u) = \begin{bmatrix} -i_E \\ -i_C \end{bmatrix} = \begin{bmatrix} I_F - \alpha_R I_R \\ -\alpha_F I_F + I_R \end{bmatrix}.$$

With

$$g_F = \frac{I_{ES}}{V_T} e^{v_{BE}/V_T}, \qquad g_R = \frac{I_{CS}}{V_T} e^{v_{BC}/V_T},$$

the derivative is

$$G(u) = \begin{bmatrix} g_F & -\alpha_R g_R \\ -\alpha_F g_F & g_R \end{bmatrix}.$$

For a physically consistent Ebers–Moll parameterization, enforce

$$\alpha_F I_{ES} = \alpha_R I_{CS}, \qquad 0 < \alpha_F, \alpha_R < 1.$$

This initial model omits Early effect, breakdown, and detailed charge storage. Document these limitations.

Use numerically stable exponential evaluation, including `expm1` near zero. Use Newton step limiting or a documented, derivative-consistent exponential continuation to prevent overflow. Do not silently clamp currents while retaining inconsistent derivatives.

6 Harmonic-balance equations

For each retained harmonic,

$$R_k = A_{\mathrm{L}}(k\omega_0) X_k + I_{\mathrm{NL},k}(X) - S_k = 0.$$

Nonlinear currents enter the node-equation rows. Their contribution to auxiliary branch-constraint rows is zero unless a particular device model explicitly introduces additional equations.

Linear components remain independent between frequencies. Nonlinear components couple frequencies.

6.1 Alternating frequency/time evaluation

For period $T = 2\pi/\omega_0$, choose

$$t_m = \frac{mT}{M}, \qquad m = 0, \ldots, M-1.$$

Reconstruct terminal voltages:

$$u_{d,m} = \sum_{\ell \in \mathcal{K}} B_d V_\ell e^{j\ell\omega_0 t_m}.$$

Evaluate

$$f_{d,m} = f_d(u_{d,m}), \qquad q_{d,m} = q_d(u_{d,m}).$$

Transform back:

$$F_{d,k} = \frac{1}{M} \sum_m f_{d,m} e^{-jk\omega_0 t_m}, \qquad Q_{d,k} = \frac{1}{M} \sum_m q_{d,m} e^{-jk\omega_0 t_m}.$$

Then

$$\boxed{I_{\mathrm{NL},k} = \sum_d B_d^T \left(F_{d,k} + jk\omega_0 Q_{d,k} \right).}$$

For sparse active spectra, zero-fill inactive coefficients on the FFT grid. Document the FFT library's normalization and index conventions.

A grid capable of reconstructing the voltage spectrum is not necessarily large enough to resolve the nonlinear current spectrum. Oversample and increase M until retained coefficients and candidate error indicators stabilize. Exponential models have no finite exactly alias-free grid.

6.2 Newton Jacobian

At every time sample evaluate $G_d(u)$ and $C_d(u)$, then transform these derivative matrices to obtain coefficients $G_{d,p}$ and $C_{d,p}$.

For a signed-frequency representation, the node-voltage blocks are

$$\boxed{J_{k\ell} = \delta_{k\ell} A_{\mathrm{L}}(k\omega_0) + \sum_d B_d^T \left(G_{d,k-\ell} + jk\omega_0 C_{d,k-\ell}\right) B_d.}$$

In this formula, embed B_d and the nonlinear stamps into the full MNA layout as necessary. The charge multiplier is the output frequency $jk\omega_0$, not the input frequency $j\ell\omega_0$.

Solve

$$J\Delta x = -R, \qquad x_{\mathrm{new}} = x + \lambda\Delta x, \qquad 0 < \lambda \leq 1.$$

For a real packed spectrum, include the dependence of negative-frequency coefficients on conjugate positive-frequency coefficients. Do not simply discard the negative-frequency Jacobian blocks.

A practical alternative is a Jacobian-vector product:

1. Reconstruct the voltage perturbation in time.
2. Compute $\delta f_m = G_m \delta u_m$ and $\delta q_m = C_m \delta u_m$.
3. Transform back and form $\delta F_k + jk\omega_0 \delta Q_k$.
4. Add the linear MNA product.

Initially use a directly assembled Jacobian for small test circuits. Introduce matrix-free Newton–Krylov methods only after validation.

6.3 Convergence and robustness

Use:

- A DC operating-point solve before AC continuation.
- Source stepping from small AC amplitude to the target amplitude.
- Damped Newton steps with a scaled residual merit function.
- Separate absolute and relative tolerances for voltage, current, and constraint equations.
- Both residual and update-size convergence checks.
- Iteration limits and explicit failure diagnostics.

MNA rows have different units. Never compare all raw residual entries using one unscaled tolerance.

Detect floating nodes, inconsistent ideal-source constraints, and singular matrices. Do not silently hide these problems with arbitrary regularization.

7 Adaptive frequency selection

7.1 Why source-only selection fails

Nonlinearities generate DC, harmonics, and mixing products. For example, v^2 produces sums and differences of input frequencies. Transistor exponentials can generate arbitrarily high orders.

Adaptation must measure the equations omitted by the active spectrum, rather than only inspect source types.

7.2 Active and candidate sets

Maintain:

- An active set $\mathcal{K}$ used by the nonlinear solve.
- A larger candidate set $\mathcal{P}$ inspected for spectral error.
- A time-sampling grid sufficiently large to inspect $\mathcal{P}$.

Always retain DC. Add and remove conjugate pairs together.

Maintain source coefficients on the candidate set even when they are not active. This matters for truncated square and sawtooth sources.

7.3 Omitted-equation residual

After convergence on $\mathcal{K}$, evaluate nonlinear currents on the larger grid. For omitted p, set $X_p = 0$ and compute

$$\boxed{r_p = I_{\mathrm{NL},p} - S_p.}$$

This residual may include nonzero voltage-source constraint entries through S_p.

Current magnitude alone is not a reliable spectral error measure: small currents near resonances can cause large voltages.

Estimate the missing solution by

$$\Delta \widehat{X}_p = -D_p^{-1} r_p,$$

where

$$D_p = A_{\mathrm{L}}(\omega_p) + \mathrm{embed}\left[\sum_d B_d^T (G_{d,0} + j\omega_p C_{d,0}) B_d\right].$$

This is an approximate incremental loading calculation. It ignores cross-frequency corrections.

For node-voltage entries define

$$\eta_p = \max_n \frac{|\Delta \widehat{V}_{n,p}|}{V_{\mathrm{abs}} + V_{\mathrm{rel}} V_{\mathrm{scale},n}}.$$

Add separate current or observable error indicators if needed. A voltage-only indicator does not bound every branch-current error.

If the estimator block is singular or ill-conditioned, do not classify the candidate as harmless. Expand the candidate solve, improve the coupled estimator, or report an unresolved error.

7.4 Adaptation loop

```
active = initialSourceHarmonicsPlusDC()
x = solveDCAndInitialize(active)

repeat:
    x = solveHarmonicBalance(active, x)
    ensureSamplingConvergence(x)

    candidates = buildGuardSpectrum(active, sources)
    residuals = evaluateOmittedEquations(x, candidates)
    indicators = estimateMissingSolutionErrors(residuals)

    additions = selectWorstCandidates(indicators)
    if additions is empty and guardSpectrumIsAdequate:
        return convergedResult

    active = active union additions
    x = warmStartWithPredictedCoefficients(x, additions)

until resourceLimit

return explicitNonconvergenceDiagnostics
```

Inspect beyond the immediate active spectrum and enlarge the guard band until its outer edge is insignificant. A small residual inside a narrow candidate set does not prove that more distant harmonics are negligible.

For removal, require:

1. Small contribution for several adaptation cycles.
2. A removal threshold lower than the addition threshold.
3. An omitted-residual test after trial removal.
4. Re-solution of the reduced system.

Never remove DC or explicitly protected source components.

Declare separate outcomes for nonlinear convergence, spectral convergence, and sampling convergence.

8 Frequency representation

8.1 Commensurate sources

For a common fundamental, use integer indices

$$\omega_k = k\omega_0.$$

Do not identify harmonics by floating-point equality.

Sparse active coefficients can coexist with a dense FFT workspace.

8.2 Incommensurate sources

For independent fundamentals $\Omega_1, \ldots, \Omega_d$, use integer tuples:

$$\omega_{\mathbf{k}} = \mathbf{k} \cdot \mathbf{\Omega}, \qquad \mathbf{k} \in \mathbb{Z}^d.$$

Examples:

$$(1,0) \mapsto \Omega_1, \quad (1,-1) \mapsto \Omega_1 - \Omega_2, \quad (2,1) \mapsto 2\Omega_1 + \Omega_2.$$

Evaluate the quasiperiodic waveform on a multidimensional phase grid. The charge multiplier becomes $j\omega_{\mathbf{k}}$. Use multidimensional FFTs or equivalent Fourier projection.

Reduce rationally dependent fundamentals to an independent basis. Otherwise different tuples can describe the same physical frequency.

Implement the one-fundamental periodic solver first. Multitone adaptation is a later milestone because its frequency lattice can grow rapidly.

9 Suggested Java architecture

Adapt names and types to the existing repository. Avoid introducing duplicate complex-number or sparse-matrix libraries unnecessarily.

```
interface LinearComponent {
    void allocateUnknowns(UnknownAllocator allocator);
    void stamp(
        double angularFrequency,
        ComplexMatrixAccumulator matrix,
        ComplexVectorAccumulator rhs,
        UnknownLayout layout);
}

interface NonlinearDevice {
    TerminalMapping mapping();

    // Currents and charges use the reduced terminal coordinates.
    void evaluate(
        double[] terminalVoltages,
        double[] currents,
        double[] charges,
        double[][] currentJacobian,
        double[][] chargeJacobian);
}

interface PeriodicSource {
    Complex coefficient(int harmonicIndex);
}

interface SpectrumTransform {
    // Enforce documented normalization and conjugate symmetry.
    void toTime(Spectrum spectrum, TimeSamples output);
    void toFrequency(TimeSamples samples, Spectrum output);
}
```

Recommended responsibilities:

UnknownLayout Ground handling, node indices, auxiliary currents, and real-spectrum packing.

LinearAssembler Frequency-dependent MNA stamps and source vectors.

DeviceEvaluator Time-domain device evaluation and derivatives.

HarmonicBalanceResidual Transform orchestration and residual assembly.

NewtonSolver Scaling, damping, convergence, and diagnostics.

SpectrumController Candidate generation, error estimation, additions, removals, and hysteresis.

SamplingController FFT grid selection and aliasing checks.

Cache linear matrices when frequency and component parameters are unchanged. Reuse work arrays and sparse symbolic factorizations where supported. Never reuse numerical factorizations after matrix entries change.

10 Implementation milestones for Copilot

Milestone 1: inspect and preserve behavior

1. Inspect the existing graph representation, ground convention, Fourier normalization, solver, and component APIs.
2. Identify tests and establish baseline results.
3. Propose a minimal migration plan before large edits.
4. Preserve public APIs where feasible.

Milestone 2: linear stamps and MNA

1. Wrap existing scalar impedances with a two-terminal stamp.
2. Add current-source and voltage-source stamps.
3. Add branch-current treatment for inductors, including DC.
4. Validate existing linear simulations before introducing nonlinear frequency coupling.

Milestone 3: DC nonlinear solver

1. Add nonlinear device current and derivative contracts.
2. Implement a diode and conductive Ebers–Moll NPN.
3. Implement damped Newton and source stepping.
4. Validate analytic derivatives against finite differences.

Milestone 4: fixed-spectrum harmonic balance

1. Implement real Fourier packing and transform tests.
2. Implement time-domain nonlinear evaluation.
3. Implement residual and coupled Jacobian assembly.
4. Solve a user-specified harmonic set.
5. Validate against enlarged fixed-spectrum solutions.

Milestone 5: adaptive spectrum

1. Add guard-spectrum evaluation and omitted residuals.
2. Add incremental missing-voltage estimation.
3. Add batch expansion and warm starts.
4. Add sampling refinement.
5. Add conservative removal only after addition is reliable.

Milestone 6: charge and advanced models

1. Add documented terminal charge models and derivatives.
2. Validate the output-frequency charge multiplier.
3. Add improved BJT or MOSFET models as separate implementations.
4. Consider multitone and matrix-free solvers after the periodic implementation is stable.

Each milestone should compile, include tests, and avoid unrelated refactoring.

11 Required verification tests

11.1 Linear assembly

- Resistor divider.
- RC and RL frequency responses.
- Voltage-source polarity and current-source direction.
- Inductor at DC.
- Controlled-source signs.
- Explicit detection of floating or inconsistent circuits.
- Agreement with the original solver for supported circuits.

11.2 Device models

- BJT terminal-current conservation.
- Finite-difference verification of every derivative entry.
- Reverse-biased and forward-biased junction behavior.
- Bias points spanning cutoff, forward active operation, and saturation.
- Consistency of charge derivatives when charge models are added.

11.3 Fourier and harmonic balance

- FFT round trip, DC normalization, and conjugate symmetry.
- For $v = A\cos(\omega_0 t)$ and $i = av^2$, verify $I_0 = aA^2/2$ and $I_{\pm 2} = aA^2/4$.
- A linear circuit solved by harmonic balance must reproduce the independent-frequency solver.
- A small nonlinear perturbation must approach the corresponding small-signal result.
- A driven nonlinear circuit must generate DC and harmonics.
- Compare Jacobian-vector products with finite-difference residual perturbations, including charge terms.
- Increasing the sampling grid must stabilize retained results.

11.4 Adaptive spectrum

- Begin with DC and the driven fundamental; verify generated harmonics are added.
- Use a resonant network to verify that a small omitted current with a large voltage effect is retained.
- Compare adaptive solutions with generously sized fixed-spectrum reference solutions.
- Verify tighter tolerances reduce observable error.
- Verify removal hysteresis prevents repeated add/remove cycles.
- Verify resource limits report unresolved spectral error rather than claiming convergence.

12 Acceptance criteria

The refactor is complete when:

1. Existing linear results remain correct.
2. Ideal voltage sources and inductors work at DC using MNA.
3. Nonlinear transistors generate DC shifts and harmonics.
4. Newton derivatives pass numerical verification.
5. Adaptive solutions agree with expanded-spectrum references within configured tolerances.
6. Aliasing checks are independent of active-set convergence.
7. Unsupported regimes and convergence failures are clearly reported.

13 Limitations to document

Harmonic balance solves an assumed periodic or quasiperiodic steady state. Convergence does not prove dynamic stability or uniqueness.

Autonomous oscillators require an unknown oscillation frequency and a phase condition. Subharmonic solutions may require a lower fundamental than the source frequency. General startup behavior and chaotic signals require time-domain simulation or other specialized methods.

Adaptive frequency indicators are practical error estimates, not a universal rigorous bound. Validate their effectiveness against larger spectra and refined sampling grids.